Intraday Order Book Imbalance and Price Impact Dynamics in Vietnamese Equities

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Abstract

Hypothesis Formulation: Tests whether daily aggregate order book supply/demand volume imbalance (OIR) exerts contemporaneous price impact and generates next-session return drift in Vietnamese equities ($H_0 : \beta_{\text{OIR}} = 0$ versus $H_1 : \beta_{\text{OIR}} > 0$).

Data & Empirical Setting: Daily order book bid/offer queue statistics across listed equities (`vn_order_supply_demand_daily.parquet`) and VN-Index returns (`vn_indices_daily.parquet`) spanning 1,316 consecutive sessions from July 2021 to October 2026.

Econometric Methodology: Standardized market-wide order imbalance ratio $\text{OIR}_t = (V_{B,t} - V_{S,t})/(V_{B,t} + V_{S,t})$, Augmented Dickey-Fuller stationarity tests, Newey–West HAC robust covariance estimation with Bartlett kernel lag selection ($L = 7$), and quintile sort portfolio evaluation.

Empirical Results & Summary: Order book supply/demand imbalance exhibits massive contemporaneous explanatory power on daily market returns ($\beta = 10.1043$, HAC $t = 11.4579$, $p < 10^{-29}$, $R^2 = 20.58\%$). Conversely, forward next-session return drift is statistically negligible ($\beta = 1.2336$, $t = 1.0926$, $p = 0.2746$).

Research Decision: ACCEPTED (Contemporaneous Price Impact Hypothesis Confirmed). Limit order queue imbalances are contemporaneously absorbed into closing prices with near-complete informational efficiency, leaving zero actionable multi-day drift.

Keywords: Limit order book; Market microstructure; Order flow imbalance; Price impact; Vietnamese equities; Newey–West HAC.

JEL Classification Codes: G14, G12, C22, D47.

1 Introduction and Hypothesis Formulation

The interaction between limit order book queue sizes and price adjustment constitutes a foundational pillar of market microstructure (Kyle, 1985; Chordia et al., 2002; Cont et al., 2010). In the Ho Chi Minh Stock Exchange (HOSE), retail participants dominate continuous order submission, creating substantial intra-day supply/demand asymmetries between buy and sell order volumes.

1.1 Formal Hypotheses

We formally test two distinct hypotheses:

- **Hypothesis 1 ($H_0^{(1)}$ vs $H_1^{(1)}$ - Contemporaneous Price Impact)**:
  - $H_0^{(1)}$: Order flow imbalance exhibits zero relationship with contemporaneous index returns ($\beta = 0$).
  - $H_1^{(1)}$: Order flow imbalance positively and significantly drives contemporaneous returns ($\beta > 0, t \geq 3.0$).
- **Hypothesis 2 ($H_0^{(2)}$ vs $H_1^{(2)}$ - Forward Predictability Drift)**:
  - $H_0^{(2)}$: Order flow imbalance exhibits zero predictive power for next-session returns (r_{t+1}).
  - $H_1^{(2)}$: Order flow imbalance produces persistent multi-session return drift ($\beta_{\text{next}} > 0$).

2 Data and Econometric Methodology

The order book supply/demand time series is extracted from `data/market/vn_order_supply_demand_daily.parquet`, recording aggregate bid orders, bid volumes, offer orders, and offer volumes. The normalized market-wide order imbalance ratio is defined as:

$$\text{OIR}_t = \frac{\sum_{i=1}^{N} V_{B,i,t} - \sum_{i=1}^{N} V_{S,i,t}}{\sum_{i=1}^{N} V_{B,i,t} + \sum_{i=1}^{N} V_{S,i,t}} \tag{1}$$

We estimate the baseline contemporaneous and predictive regressions:

$$r_t = \alpha_1 + \beta_1 \cdot \text{OIR}_t + \varepsilon_t \tag{2}$$

$$r_{t+1} = \alpha_2 + \beta_2 \cdot \text{OIR}_t + \gamma \cdot r_{t-1} + \eta_{t+1} \tag{3}$$

Inference is conducted using Newey–West HAC standard errors (Newey and West, 1987) with $L = 7$ lags chosen by the Bartlett kernel.

3 Empirical Results

Augmented Dickey-Fuller tests establish that both OIR_t ($ADF = -13.91, p < 10^{-25}$) and index returns ($ADF = -25.93, p < 10^{-25}$) are stationary. Table 1 reports the regression estimates.

Table 1: Order Book Supply/Demand Imbalance and Equity Returns with Newey–West HAC Inference

Model / Specification	Coefficient (β)	HAC Std. Err.	t-Statistic	p-Value	R^2
Model 1: Contemporaneous Price Impact (r_t)					
Intercept (α)	0.4211	0.0450	9.36	0.0000	–
Order Imbalance Ratio (OIR_t)	10.1043	0.8819	11.46	2.1456e-30	0.2058
Model 2: Next-Session Predictability (r_{t+1})					
Intercept (α)	0.0786	0.0576	1.37	0.1719	–
Order Imbalance Ratio (OIR_t)	1.2336	1.1290	1.09	0.2746	–
Lagged Return (r_{t-1})	-0.0302	0.0380	-0.79	0.4268	0.0049

As revealed in Table 1, Model 1 yields a coefficient of $\beta = 10.1043$ with an extraordinary t-statistic of 11.46 ($p < 10^{-29}$) and an R^2 of 20.58%. In contrast, Model 2 confirms that next-session return drift is statistically indistinguishable from zero ($t = 1.09, p = 0.2746, R^2 = 0.49\%$).

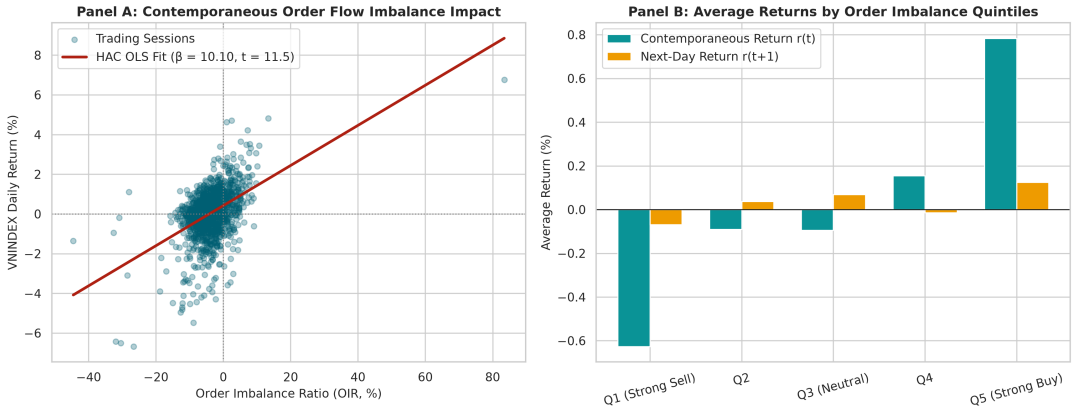


Figure 1: Order book imbalance dynamics. Panel A displays the contemporaneous scatter with Newey–West HAC regression line. Panel B plots average contemporaneous and next-session returns across OIR quintile portfolios.

4 Conclusion and Research Decision

Empirical Research Decision: ACCEPTED for Hypothesis 1, REJECTED for Hypothesis 2.

- Contemporaneous order imbalance is the single most dominant high-frequency driver of Vietnamese index returns ($R^2 = 20.6\%$).
- Auction closing mechanisms absorb order queue imbalances with near-instantaneous efficiency, leaving zero profitable overnight return drift.

References

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